

Sacha Durain

PhD Candidate – Structural Dynamics & Scientific Machine Learning

Available from September 2026
sacha.durain@gmail.com
+33 6 35 02 05 94
linkedin.com/in/sacha-durain
sdurain.github.io

PhD candidate in structural dynamics and Scientific Machine Learning, developing interpretable and uncertainty-aware predictive models from experimental testing, finite element simulations, and dynamic signals. Background spanning mechanical modeling, multimodal data analysis, surrogate modeling, and industrial simulation workflows.

Research profile

- Scientific Machine Learning for mechanics, with emphasis on surrogate modeling, uncertainty quantification, explainability, and digital twins.
- Friction-induced vibration and brake squeal prediction using coupled experimental and numerical data.
- Multimodal pipelines combining instrumented tests, finite element simulations, time-series signals, and microscopy images.

Core skills

Machine & deep learning: supervised/unsupervised learning, computer vision, uncertainty quantification, PyTorch, scikit-learn

Numerical simulation: finite element methods, Ansys, Abaqus, HyperMesh

Data analysis & visualization: NumPy, pandas, Matplotlib, Plotly

Programming: Python, MATLAB, Bash, HPC, Slurm, Git

Languages: French (native), English (C1)

Research experience

PhD Candidate

2023 – present

LaMcube & LAMIH | Lille & Valenciennes, France

• *Thesis objective: predictive and explainable models for brake squeal using experimental and numerical data.*

- Acquired, structured, and processed instrumented test data from multimodal sensors and numerical simulations.
- Developed surrogate models for a braking-system digital twin: CNNs, MLPs, Gaussian processes, recurrent models, deep ensembles, Monte Carlo methods, and SHAP-based interpretability.
- Built an open-source semantic-segmentation pipeline for brake pad material images (SEM and optical), using pseudo-labeling, unsupervised learning, and transfer learning.
- Disseminated the work through three scientific presentations and several published or submitted journal articles.

Teaching Assistant

2023 – present

Polytech Lille & INSA Hauts-de-France | Lille & Valenciennes, France

- *Dynamics & signal processing:* taught system dynamics and fundamentals of signal processing.
- *MATLAB programming:* supervised student projects in finite element simulation, optimization, and data analysis.

Industry experience

Simulation & Design Engineer – Apprentice

2020 – 2023

Alstom | Petite-Forêt, France

- Designed and validated railway products through mechanical simulation and data analysis.
- Collaborated with multidisciplinary and international teams across several sites.

Mechanical Engineering Intern

May 2021 – August 2021

Canray | Bursa, Türkiye

- Conducted test-simulation correlations on automotive components.
- Supported quality monitoring and technical exchanges with the client.

Education

PhD – Structural Dynamics & Scientific Machine Learning

2023 – present

University of Lille | Lille, France

- Dissertation: “Friction-induced vibrations: experiments and modeling of triggering conditions through enriched interface parameters.” Supervised by Prof. P. Dufrénoy and Prof. M. Massa.

Engineering Degree – Mechanical Simulation

2018 – 2023

INSA Hauts-de-France | Valenciennes, France

- Core coursework: optimization, finite element methods, programming, and dynamics.

Publications and conference presentations

Publications

1. Quentin Caradec, Sacha Durain, Maël Thévenot, Mathis Briatte, Merten Stender, Jean-François Brunel, Philippe Dufrénoy
Pin-on-Disc Experimental Study of Thermomechanical Processes Related to Squeal Occurrence.
Lubricants – published.
2. Sacha Durain, Quentin Caradec, Mathis Briatte, Jean-François Brunel, Cédric Hubert, Franck Massa, Philippe Dufrénoy
Enhancing the prediction of squeal through multiphysics and multiscale analysis of experimental data from a pin-on-disk system.
Mechanical Systems and Signal Processing (MSSP) – published.
3. Sacha Durain, Quentin Caradec, Mathis Briatte, Jean-François Brunel, Cédric Hubert, Franck Massa, Philippe Dufrénoy
Experimental-numerical validation of a contact-interface-induced squeal model under uncertainty.
Mechanical Systems and Signal Processing (MSSP) – submitted.
4. Sacha Durain, Quentin Caradec, Mathis Briatte, Jean-François Brunel, Cédric Hubert, Franck Massa, Philippe Dufrénoy
A numerical framework for long-time friction-induced vibration simulations based on an enhanced contact description.
Journal of Sound and Vibration (JSV) – submitted.

Conference presentations

1. **European Solid Mechanics Conference (ESMC 2025)**
Lyon, France, 7–11 July 2025
Quentin Caradec, Sacha Durain, Maël Thévenot, Mathis Briatte, Merten Stender, Jean-François Brunel, Philippe Dufrénoy
Experimental and numerical study of the thermomechanical and tribological mechanisms involved in dry friction brake emissions.
2. **26th French Congress of Mechanics (CFM 2025)**
Metz, France, 25–29 August 2025
Sacha Durain, Quentin Caradec, Mathis Briatte, Jean-François Brunel, Cédric Hubert, Franck Massa, Philippe Dufrénoy
Enrichment of the contact-interface description for the identification of friction-induced dynamic instabilities.
3. **CSMA 2026 – 17th National Colloquium on Structural Computation**
Presqu'île de Giens, Var, France, 18–22 May 2026
Sacha Durain, Quentin Caradec, Mathis Briatte, Jean-François Brunel, Cédric Hubert, Franck Massa, Philippe Dufrénoy
Enhanced experimental-numerical correlation for the investigation of dynamic instabilities under uncertain frictional interface modeling.